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| 1    | 2                                                                                                                                                                                                                            | 3        | 4    | 5          | 6          | 7                                  | 8 |                        |                   |             |                                                 |                   |                           |       |
| A    | <div>UniSec</div> <div>Medium Voltage Switchgear</div> <div>PT MULTIMAS NABATI ASAHAN</div> <div>MVD-25 &amp; MVD-26 20kV Centralized Panel Room MNA</div> <div>SCHEMATIC DIAGRAM</div> <div>DRC BUS PT</div> <div>H07</div> |          |      |            |            |                                    |   | A                      |                   |             |                                                 |                   |                           |       |
| B    |                                                                                                                                                                                                                              |          |      |            |            |                                    |   | B                      |                   |             |                                                 |                   |                           |       |
| C    |                                                                                                                                                                                                                              |          |      |            |            |                                    |   | C                      |                   |             |                                                 |                   |                           |       |
| D    |                                                                                                                                                                                                                              |          |      |            |            |                                    |   | D                      |                   |             |                                                 |                   |                           |       |
| E    |                                                                                                                                                                                                                              |          |      |            |            |                                    |   | E                      |                   |             |                                                 |                   |                           |       |
| F    |                                                                                                                                                                                                                              |          |      |            |            |                                    |   | F                      |                   |             |                                                 |                   |                           |       |
|      |                                                                                                                                                                                                                              |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE                      |   | ABB                    | EPMV              | COVER SHEET | Scale                                           |                   | = H07                     |       |
|      |                                                                                                                                                                                                                              |          |      | Prepared   | YR         | MV Switchgear for MNA Serang Plant |   |                        |                   |             | MVD-25 & MVD-26 20kV Centralized Panel Room MNA | Customer          | PT Multimas Nabati Asahan | & 308 |
| 00   | Issued for Approval                                                                                                                                                                                                          | 15.01.19 | YR   | Checked    | AW         |                                    |   | PT. ABB Sakti Industri | SCHEMATIC DIAGRAM | DRC BUS PT  | Drawing Nbr.                                    | 18.042-MNA-308.07 | Sh.                       | 1     |
| Ind. | Revision                                                                                                                                                                                                                     | Date     | Name | Approved   | AI         | Based on: BE318042                 |   |                        |                   |             | Customer Doc. Nbr.                              |                   | Cont.                     | 16    |
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| A | SHEET INDEX                |                     |                                            |      |            |            |                                    |   | A                      |      |                                                 |  |                    |                           |       |    |
| B | PAGE                       |                     | PAGE NAME                                  |      |            | REVISION   |                                    |   | B                      |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/1  |                     | COVER SHEET                                |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/2  |                     | Index of Sheet                             |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/3  |                     | REFERENCE DESIGNATIONS                     |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/4  |                     | REFERENCE DESIGNATIONS                     |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/5  |                     | REFERENCE DESIGNATIONS                     |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/6  |                     | REFERENCE DESIGNATIONS                     |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/7  |                     | DISTRIBUTION OF AUXILIARY CIRCUITS         |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/8  |                     | MAIN CIRCUITS                              |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
|   | ==MVD-25_MVD-26=H07&308/9  |                     | LV DOOR LAYOUT                             |      |            | 00         |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
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|   | ==MVD-25_MVD-26=H07&308/12 |                     | Terminal diagram ==MVD-25_MVD-26=H07+-XDI  |      |            |            |                                    |   |                        |      |                                                 |  |                    |                           |       |    |
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| C | ==MVD-25_MVD-26=H07&308/16 |                     | Terminal diagram ==MVD-25_MVD-26=H07+-XDX  |      |            |            |                                    |   | C                      |      |                                                 |  |                    |                           |       |    |
| D |                            |                     |                                            |      |            |            |                                    |   | D                      |      |                                                 |  |                    |                           |       |    |
| E |                            |                     |                                            |      |            |            |                                    |   | E                      |      |                                                 |  |                    |                           |       |    |
| F |                            |                     |                                            |      | Date Prep. | 15.11.2018 | PROJECT TITLE                      |   | ABB                    | EPMV | Index of Sheet                                  |  | Scale              |                           | = H07 | F  |
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|---|---------------|----------------------------------------------------------------------------------------------------------------------|----------|------|------------|------------|-----------------------------------------------------|----------------|--------------------------------------------------------------------------------------------|------|-------------------------------------------------|--|--------------|---------------------------|-------|---|
| A | -BGI31        | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION                                                     |          |      |            |            |                                                     | -CA            | CAPACITORS                                                                                 |      |                                                 |  |              |                           | A     |   |
|   | -BGI32        | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION                                                   |          |      |            |            |                                                     | -CA1           | CAPACITOR FOR FRONT VENTILATOR                                                             |      |                                                 |  |              |                           |       |   |
|   | -BGI33        | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED) |          |      |            |            |                                                     | -CA2           | CAPACITOR FOR REAR VENTILATOR                                                              |      |                                                 |  |              |                           |       |   |
|   | -BGI41...     | POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION                                      |          |      |            |            |                                                     | -EA1           | LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.                                      |      |                                                 |  |              |                           |       |   |
|   | -BGI42        | POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION                                               |          |      |            |            |                                                     | -EA2           | LIGHTING LAMP LOCATED IN CABLE COMPARTMENT                                                 |      |                                                 |  |              |                           |       |   |
|   | -BGI43...     | POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION                                       |          |      |            |            |                                                     | -EA3           | LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE                                     |      |                                                 |  |              |                           |       |   |
|   | -BGI51        | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION                                            |          |      |            |            |                                                     | -EA4           | LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT                                      |      |                                                 |  |              |                           |       |   |
|   | -BGI52        | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION                                                     |          |      |            |            |                                                     | -EB1           | HEATER LOCATED IN CABLE COMPARTMENT                                                        |      |                                                 |  |              |                           |       |   |
|   | -BGI53        | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION                                             |          |      |            |            |                                                     | -EB2           | HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT                                              |      |                                                 |  |              |                           |       |   |
|   | -BGK          | POSITION SWITCH OF KEY LOCK                                                                                          |          |      |            |            |                                                     | -EB3           | HEATER LOCATED IN MOTOR CABINET                                                            |      |                                                 |  |              |                           |       |   |
| B | -BGL1         | POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1                                                                      |          |      |            |            |                                                     | -EB4           | HEATER FOR VENTILATION TEST                                                                |      |                                                 |  |              |                           | B     |   |
|   | -BGS1...-BGS4 | POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS                                                                         |          |      |            |            |                                                     | -EB5           | HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.                                              |      |                                                 |  |              |                           |       |   |
|   | -BGS5         | PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION                                                              |          |      |            |            |                                                     | -EB7           | HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE                          |      |                                                 |  |              |                           |       |   |
|   | -BGT1         | POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION                                                      |          |      |            |            |                                                     | -EB8           | HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE                         |      |                                                 |  |              |                           |       |   |
|   | -BGT2         | POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION                                                         |          |      |            |            |                                                     | -FA1           | SURGE ARRESTER LOCATED ON PHASE L1                                                         |      |                                                 |  |              |                           |       |   |
|   | -BGT3         | POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION                                           |          |      |            |            |                                                     | -FA2           | SURGE ARRESTER LOCATED ON PHASE L2                                                         |      |                                                 |  |              |                           |       |   |
|   | -BGT4         | POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION                                                 |          |      |            |            |                                                     | -FA3           | SURGE ARRESTER LOCATED ON PHASE L3                                                         |      |                                                 |  |              |                           |       |   |
|   | -BGT5         | POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION                                                    |          |      |            |            |                                                     | -FCD           | FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS                                    |      |                                                 |  |              |                           |       |   |
|   | -BGT6         | POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION                                                          |          |      |            |            |                                                     | -FCF1          | MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1                                                    |      |                                                 |  |              |                           |       |   |
|   | -BGT7         | POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION                                                             |          |      |            |            |                                                     | -FCF2          | MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2                                                    |      |                                                 |  |              |                           |       |   |
| C | -BGT8         | POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION                                                      |          |      |            |            |                                                     | -FCF3          | MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3                                                    |      |                                                 |  |              |                           | C     |   |
|   | -BGT11        | PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION                                                                |          |      |            |            |                                                     | -FCM1          | MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER |      |                                                 |  |              |                           |       |   |
|   | -BGT12        | PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION                                                                   |          |      |            |            |                                                     | -FCM2...-FCM10 | MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS                            |      |                                                 |  |              |                           |       |   |
|   | -BGV1         | POSITION SWITCH OF FRONT VENTILATOR                                                                                  |          |      |            |            |                                                     | -FCM11         | MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1                                |      |                                                 |  |              |                           |       |   |
|   | -BGV2         | POSITION SWITCH OF REAR VENTILATOR                                                                                   |          |      |            |            |                                                     | -FCM12         | MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2                                |      |                                                 |  |              |                           |       |   |
|   | -BM           | HYGROSTAT                                                                                                            |          |      |            |            |                                                     | -FCM13         | MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3                                |      |                                                 |  |              |                           |       |   |
|   | -BPA4         | PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING                                        |          |      |            |            |                                                     | -FCT           | THERMAL OVERLOAD RELEASES                                                                  |      |                                                 |  |              |                           |       |   |
|   | -BPA5         | PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING               |          |      |            |            |                                                     | -GA            | GENERATORS                                                                                 |      |                                                 |  |              |                           |       |   |
|   | -BPA6         | PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING      |          |      |            |            |                                                     | -GQ1           | FRONT VENTILATOR                                                                           |      |                                                 |  |              |                           |       |   |
|   | -BPA7         | PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING                |          |      |            |            |                                                     | -GQ2           | REAR VENTILATOR                                                                            |      |                                                 |  |              |                           |       |   |
| D | -BPS1         | PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER                                 |          |      |            |            |                                                     | -GQ3           | VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK                                                |      |                                                 |  |              |                           | D     |   |
|   | -BPS2         | PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2                                                          |          |      |            |            |                                                     | -KFA1          | AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE                                                |      |                                                 |  |              |                           |       |   |
|   | -BPS3         | PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3                                                          |          |      |            |            |                                                     | -KFA2          | AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE                                       |      |                                                 |  |              |                           |       |   |
|   | -BR           | FLAME DETECTORS, SMOKE DETECTORS                                                                                     |          |      |            |            |                                                     | -KFA3...-KFA9  | AUXILIARY RELAYS OR CONTACTORS                                                             |      |                                                 |  |              |                           |       |   |
|   | -BT           | THERMOSTAT                                                                                                           |          |      |            |            |                                                     | -KFC           | CLOSING RELAYS OR CONTACTORS                                                               |      |                                                 |  |              |                           |       |   |
|   | -BT1          | THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS                                                                        |          |      |            |            |                                                     | -KFI           | INTEGRATED CIRCUITS                                                                        |      |                                                 |  |              |                           |       |   |
|   | -BT2          | THERMOSTAT FOR ENERGIZATION OF VENTILATORS                                                                           |          |      |            |            |                                                     | -KFL           | LOCKOUT RELAY                                                                              |      |                                                 |  |              |                           |       |   |
|   | -BT3          | THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM                                                                         |          |      |            |            |                                                     | -KFM           | MICROPROCESSORS                                                                            |      |                                                 |  |              |                           |       |   |
|   | -BU           | BUCHHOLZ RELAY                                                                                                       |          |      |            |            |                                                     | -KFN           | ANTIPUMPING RELAYS                                                                         |      |                                                 |  |              |                           |       |   |
|   | -BX1          | UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING                                                                   |          |      |            |            |                                                     | -KFO           | OPENING RELAYS OR CONTACTORS                                                               |      |                                                 |  |              |                           |       |   |
| E | -BX2          | ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING                                                     |          |      |            |            |                                                     | -KFP           | PROGRAMMABLE LOGIC CONTROLLERS (PLC)                                                       |      |                                                 |  |              |                           | E     |   |
|   | -BZ,-BZ4      | MULTIFUNCTION SENSORS                                                                                                |          |      |            |            |                                                     | -KFR           | RECLOSING RELAYS                                                                           |      |                                                 |  |              |                           |       |   |
|   | -BZ1          | COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1                                                             |          |      |            |            |                                                     | -KFS           | SYNCHRONIZING DEVICES                                                                      |      |                                                 |  |              |                           |       |   |
|   | -BZ2          | COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2                                                             |          |      |            |            |                                                     | -KFT           | AUXILIARY TIME RELAYS, DELAY ELEMENTS                                                      |      |                                                 |  |              |                           |       |   |
|   | -BZ3          | COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3                                                             |          |      |            |            |                                                     | -KFZ           | CONTROLLERS                                                                                |      |                                                 |  |              |                           |       |   |
|   |               |                                                                                                                      |          |      |            |            |                                                     | -MBC           | CLOSING RELEASE OF CIRCUIT-BREAKER                                                         |      |                                                 |  |              |                           |       |   |
|   |               |                                                                                                                      |          |      |            |            |                                                     | -MAD           | MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD                                        |      |                                                 |  |              |                           |       |   |
|   |               |                                                                                                                      |          |      |            |            |                                                     | -MAD1          | MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1                                        |      |                                                 |  |              |                           |       |   |
|   |               |                                                                                                                      |          |      |            |            |                                                     | -MAD2          | MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2                                        |      |                                                 |  |              |                           |       |   |
|   |               |                                                                                                                      |          |      |            |            |                                                     | -MAD9          | MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS                                 |      |                                                 |  |              |                           |       |   |
| F |               |                                                                                                                      |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |                | ABB                                                                                        | EPMV | REFERENCE DESIGNATIONS                          |  | Scale        |                           | = H07 | F |
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|---|-------|----------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------|
| A | -MAE  | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE                                                         | -QBH   | MANUAL CIRCUIT-BREAKERS                                                                                      |
|   | -MAE1 | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2                                               | -QBL   | LINKS                                                                                                        |
| B | -MAE8 | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS         | -QBM   | MINIATURE SWITCH-DISCONNECTORS                                                                               |
|   | -MAS  | MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING                                                                     | -QBS   | SWITCH-DISCONNECTORS                                                                                         |
| C | -MAT  | MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT                                                         | -QBT   | TRUCK                                                                                                        |
|   | -MBC  | CLOSING RELEASE OF CIRCUIT-BREAKER                                                                             | -QCA   | EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING                                                           |
| D | -MBC4 | CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS                                                                    | -QCE   | EARTHING SWITCH                                                                                              |
|   | -MBO1 | FIRST OPENING RELEASE OF CIRCUIT-BREAKER                                                                       | -QCE1  | EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)                 |
| E | -MBO2 | SECOND OPENING RELEASE OF CIRCUIT-BREAKER                                                                      | -QCE2  | EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)                                                       |
|   | -MBO3 | OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER | -QQ    | CLUTCHES                                                                                                     |
| F | -MBO4 | OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS                                                                    | -RAA   | ANTIFERRORESONANCE RESISTOR                                                                                  |
|   | -MBU  | UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER                                                                        | -RAD   | DIODES                                                                                                       |
| G | -PFB  | BLUE SIGNAL LAMPS                                                                                              | -RAI   | INDUCTORS                                                                                                    |
|   | -PFF  | FLAG RELAYS                                                                                                    | -RAR   | RESISTORS                                                                                                    |
| H | -PFG  | GREEN SIGNAL LAMPS                                                                                             | -RF    | FILTERS                                                                                                      |
|   | -PFR  | RED SIGNAL LAMPS                                                                                               | -RLE1  | ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING                                                    |
| I | -PFS  | SHORT CIRCUIT INDICATORS                                                                                       | -RLE2  | ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT                                                       |
|   | -PFV  | VOLTAGE INDICATORS                                                                                             | -RLE3  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE           |
| J | -PFV1 | VOLTAGE INDICATORS ON FEEDER SIDE                                                                              | -RLE4  | ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION                                                 |
|   | -PFV2 | VOLTAGE INDICATORS ON BUSBAR SIDE                                                                              | -RLE5  | ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD                                             |
| K | -PFW  | WHITE SIGNAL LAMPS                                                                                             | -RLE6  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1             |
|   | -PFX  | CROSS INDICATORS, ELECTROMECHANICAL INDICATORS                                                                 | -RLE7  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2             |
| L | -PFY  | YELLOW SIGNAL LAMPS                                                                                            | -RLE8  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 |
|   | -PGA  | AMMETERS                                                                                                       | -RLE9  | ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS                                      |
| M | -PGC  | COUNTERS                                                                                                       | -RLE11 | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4                     |
|   | -PGF  | FREQUENCYMETERS                                                                                                | -RLE12 | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5                     |
| N | -PGH  | HOURMETERS                                                                                                     | -SFA   | AMMETRIC SWITCHES                                                                                            |
|   | -PGI  | PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE                                                           | -SFC   | CONTROL SWITCHES, CLOSING PUSH-BUTTONS                                                                       |
| O | -PGI1 | HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)                          | -SFL   | LOCKING CONTACTS                                                                                             |
|   | -PGI2 | HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)                              | -SFO   | OPENING PUSH-BUTTONS                                                                                         |
| P | -PGJ  | ACTIVE ENERGY METERS                                                                                           | -SFR   | RESET PUSH-BUTTONS                                                                                           |
|   | -PGK  | REACTIVE ENERGY METERS                                                                                         | -SFS   | SELECTOR SWITCHES                                                                                            |
| Q | -PGM  | MULTIFUNCTION INDICATORS                                                                                       | -SFT   | TEST PUSH-BUTTONS                                                                                            |
|   | -PGP  | POWER-FACTOR METERS                                                                                            | -SFU   | UNLOCKING PUSH-BUTTONS                                                                                       |
| R | -PGQ  | VARMETERS                                                                                                      | -SFV   | VOLTMETRIC SWITCHES                                                                                          |
|   | -PGS  | SYNCHRONOSCOPES                                                                                                | -TA    | POWER TRANSFORMERS                                                                                           |
| S | -PGV  | VOLTMETERS                                                                                                     | -TB    | CONVERTER                                                                                                    |
|   | -PGW  | WATTMETERS                                                                                                     | -TB1   | RECTIFIER FOR FIRST OPENING RELEASE                                                                          |
| T | -PJ   | ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)                                                                      | -TB2   | RECTIFIER FOR SECOND OPENING RELEASE                                                                         |
|   | -QAB  | CIRCUIT-BREAKERS                                                                                               | -TB3   | RECTIFIER FOR CLOSING RELEASE                                                                                |
| U | -QAC  | CONTACTORS (FOR POWER)                                                                                         | -TB4   | RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING                                      |
|   | -QBD  | DISCONNECTORS                                                                                                  | -TB5   | RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT                                         |
| V | -QBD1 | FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                              | -TB6   | RECTIFIER FOR UNDERVOLTAGE RELEASE                                                                           |
|   | -QBD2 | REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                               | -TB7   | ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER                                               |
| W | -QBD3 | FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                                    |        |                                                                                                              |
|   | -QBD4 | TWO-WAY DISCONNECTOR                                                                                           |        |                                                                                                              |
| X | -QBD5 | TWO-WAY DISCONNECTOR ON BUSBAR RISER                                                                           |        |                                                                                                              |
|   |       |                                                                                                                |        |                                                                                                              |

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|------|---------------------|----------|------|------------|------------|------------------------------------|--|------------------------|------|------------------------|-------------------------------------------------|-------------------|---------------------------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE                      |  | ABB                    | EPMV | REFERENCE DESIGNATIONS | Scale                                           |                   | = H07                     |
|      |                     |          |      | Prepared   | YR         | MV Switchgear for MNA Serang Plant |  |                        |      |                        | MVD-25 & MVD-26 20kV Centralized Panel Room MNA | Customer          | PT Multimas Nabati Asahan |
| 00   | Issued for Approval | 15.01.19 | YR   | Checked    | AW         |                                    |  | PT. ABB Sakti Industri |      | SCHEMATIC DIAGRAM      | Drawing Nbr.                                    | 18.042-MNA-308.07 | Sh. 5                     |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on: BE318042                 |  |                        |      | DRC BUS PT             | Customer Doc. Nbr.                              |                   | Cont. 16                  |

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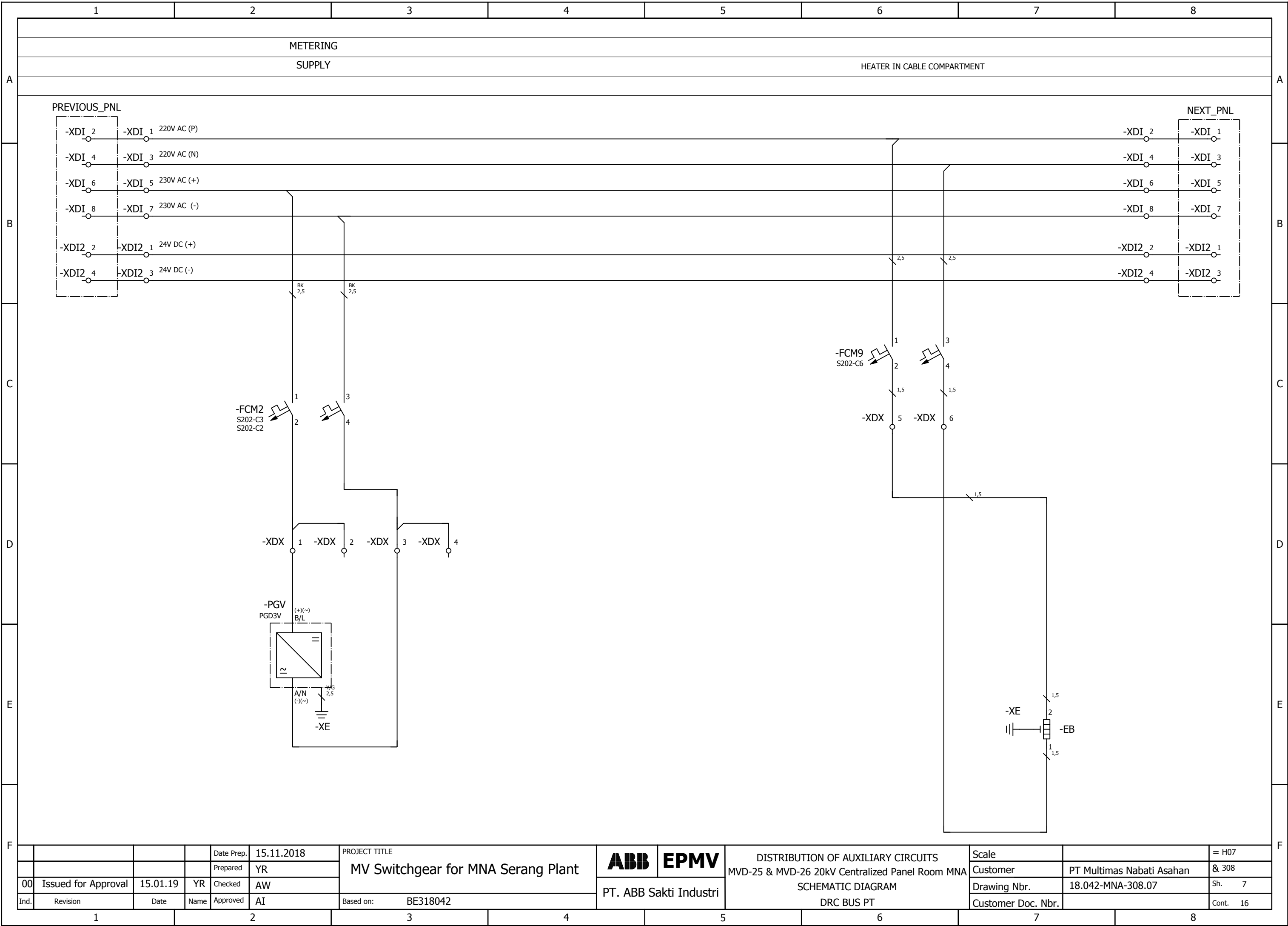
|       |                 |                                                                                                    |                                                                             |                                                                                               |   |
|-------|-----------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---|
| A     | -TFA            | ACTIVE POWER TRANSDUCERS                                                                           | -XD11                                                                       | TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS                                       | A |
|       | -TFC            | CURRENT TRANSDUCERS                                                                                | -XD12                                                                       | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING |   |
|       | -TFF            | FREQUENCY TRANSDUCERS                                                                              | -XD13                                                                       | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION    |   |
|       | -TFJ            | ACTIVE ENERGY TRANSDUCERS                                                                          | -XD14                                                                       | TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS                                |   |
|       | -TFK            | REACTIVE ENERGY TRANSDUCERS                                                                        | -XD16                                                                       | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS            |   |
|       | -TFM            | MULTIFUNCTION TRANSDUCERS                                                                          |                                                                             |                                                                                               |   |
|       | -TFP            | POWER-FACTOR TRANSDUCERS                                                                           |                                                                             |                                                                                               |   |
|       | -TFQ            | REACTIVE POWER TRANSDUCERS                                                                         |                                                                             |                                                                                               |   |
|       | -TFS            | SIGNAL CONVERTERS                                                                                  |                                                                             |                                                                                               |   |
|       | -TFV            | VOLTAGE TRANSDUCERS                                                                                |                                                                             |                                                                                               |   |
| B     | -WA             | BUSBARS                                                                                            | -XD7                                                                        | TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS                                                 | B |
|       | -WBC            | POWER CABLES                                                                                       | -XD8                                                                        | TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS                                       |   |
|       | -WE             | EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.                                                   | -XD9                                                                        | TERMINAL BLOCK FOR SPECIAL USE                                                                |   |
|       | -WGC            | CONTROL CABLES                                                                                     | -XDJ                                                                        | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD                                   |   |
|       | -WGS            | COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)                                                      | -XDJ1                                                                       | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS                             |   |
|       | -WH             | OPTICAL FIBRES                                                                                     | -XDM                                                                        | SEALABLE TERMINAL BLOCK FOR MEASUREMENT                                                       |   |
|       | -XDA            | TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS                                                | -XDS                                                                        | CURRENT SOCKETS                                                                               |   |
|       | -XDA1           | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1                                 | -XDT                                                                        | TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB                           |   |
|       | -XDA2           | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2                                 | -XDU                                                                        | CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK                        |   |
|       | -XDA3           | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3                                 | -XDU1                                                                       | TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK                                |   |
| C     | -XDA5           | CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER                                   | -XDV                                                                        | TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS                                           | C |
|       | -XDA8           | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1                                 | -XDV1                                                                       | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1                            |   |
|       | -XDA9           | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2                                 | -XDV2                                                                       | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2                            |   |
|       | -XDA10          | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3                                 | -XDV3                                                                       | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3                            |   |
|       | -XDA12          | CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER                                   | -XDV4                                                                       | CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR                                         |   |
|       | -XDB            | CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS | -XDX                                                                        | SUPPORT TERMINAL BLOCK                                                                        |   |
|       | -XDB1           | CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)           | -XE1                                                                        | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE                                        |   |
|       | -XDB2           | CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER                                     | -XE2                                                                        | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE                                       |   |
|       | -XDB10...-XDB89 | CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS                                                    | -XE5                                                                        | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE                           |   |
|       | -XDB7           | CONNECTOR FOR TRUCK POSITION CONTACTS                                                              | -XE6                                                                        | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE                                 |   |
| D     | -XDB91          | CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      | -XE7                                                                        | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE                                        | D |
|       | -XDB92          | CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      | -XE8                                                                        | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE                                |   |
|       | -XDB93          | CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR                                             | -XE9                                                                        | EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT                                |   |
|       | -XDB95          | CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET                                           |                                                                             |                                                                                               |   |
|       | -XDC            | CUSTOMER TERMINAL BLOCK                                                                            |                                                                             |                                                                                               |   |
|       | -XDC1           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER                                  |                                                                             |                                                                                               |   |
|       | -XDC2           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR                              |                                                                             |                                                                                               |   |
|       | -XDC3           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT          |                                                                             |                                                                                               |   |
|       | -XDC4           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS                             |                                                                             |                                                                                               |   |
|       | E               | -XDC5                                                                                              | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER |                                                                                               |   |
| -XDE  |                 | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE                                      |                                                                             |                                                                                               |   |
| -XDE1 |                 | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE                                       |                                                                             |                                                                                               |   |
| -XDE2 |                 | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                            |                                                                             |                                                                                               |   |
| -XDE3 |                 | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                             |                                                                             |                                                                                               |   |
| -XDF  |                 | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS                                |                                                                             |                                                                                               |   |
| -XDF1 |                 | CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR                                            |                                                                             |                                                                                               |   |
| -XDF2 |                 | CONNECTOR FOR REAR VENTILATOR CIRCUITS                                                             |                                                                             |                                                                                               |   |
| -XDF3 |                 | CONNECTOR FOR FRONT VENTILATOR CIRCUITS                                                            |                                                                             |                                                                                               |   |
| -XDH  |                 | TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)   |                                                                             |                                                                                               |   |
| F     | -XDI            | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)                                     |                                                                             |                                                                                               | F |
|       |                 |                                                                                                    |                                                                             |                                                                                               |   |
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|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE                      |  | ABB                    | EPMV | REFERENCE DESIGNATIONS | Scale                                           |                   | = H07                     |
|      |                     |          |      | Prepared   | YR         | MV Switchgear for MNA Serang Plant |  |                        |      |                        | MVD-25 & MVD-26 20kV Centralized Panel Room MNA | Customer          | PT Multimas Nabati Asahan |
| 00   | Issued for Approval | 15.01.19 | YR   | Checked    | AW         |                                    |  | PT. ABB Sakti Industri |      | SCHEMATIC DIAGRAM      | Drawing Nbr.                                    | 18.042-MNA-308.07 | Sh. 6                     |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on: BE318042                 |  |                        |      | DRC BUS PT             | Customer Doc. Nbr.                              |                   | Cont. 16                  |

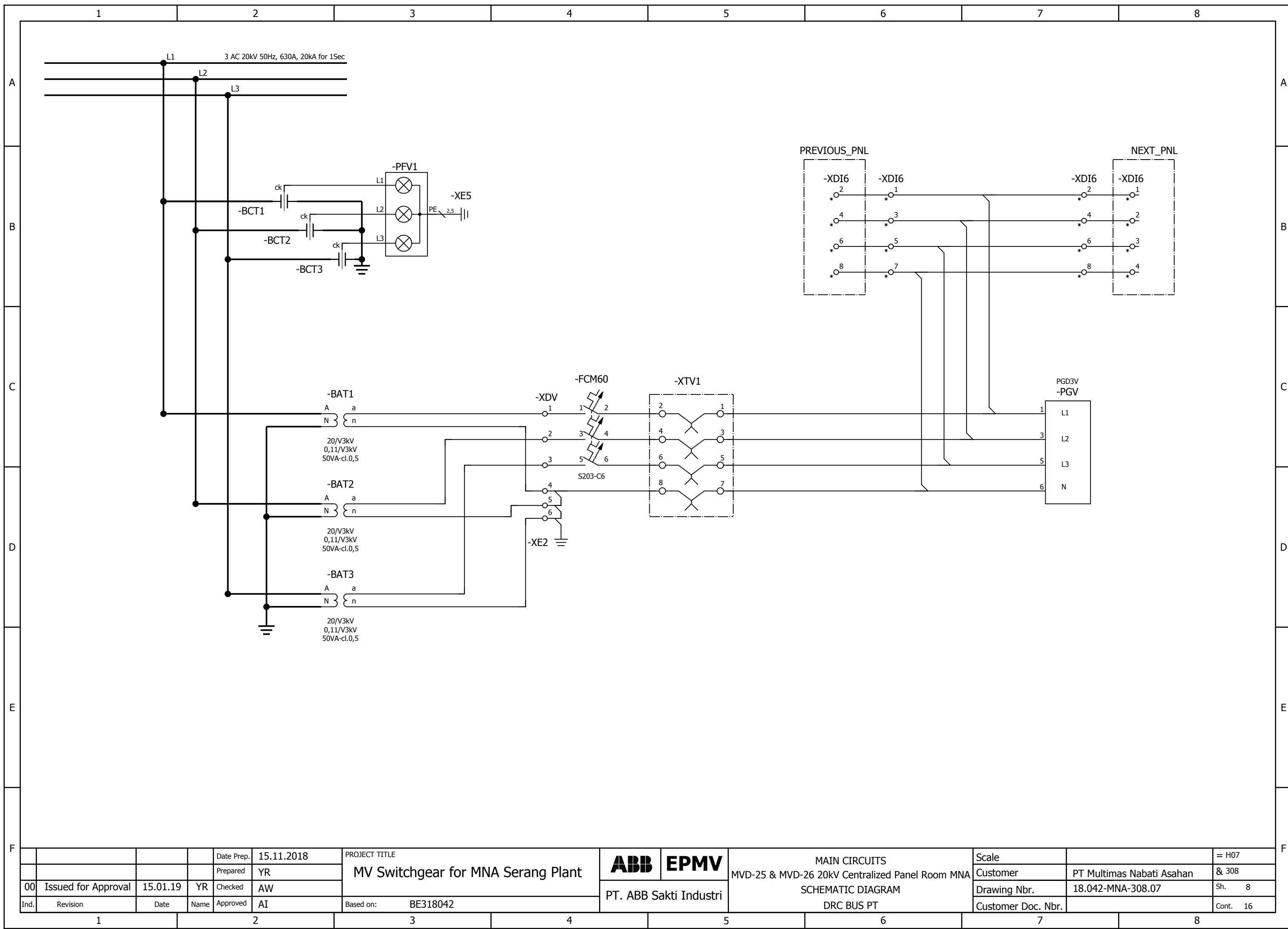
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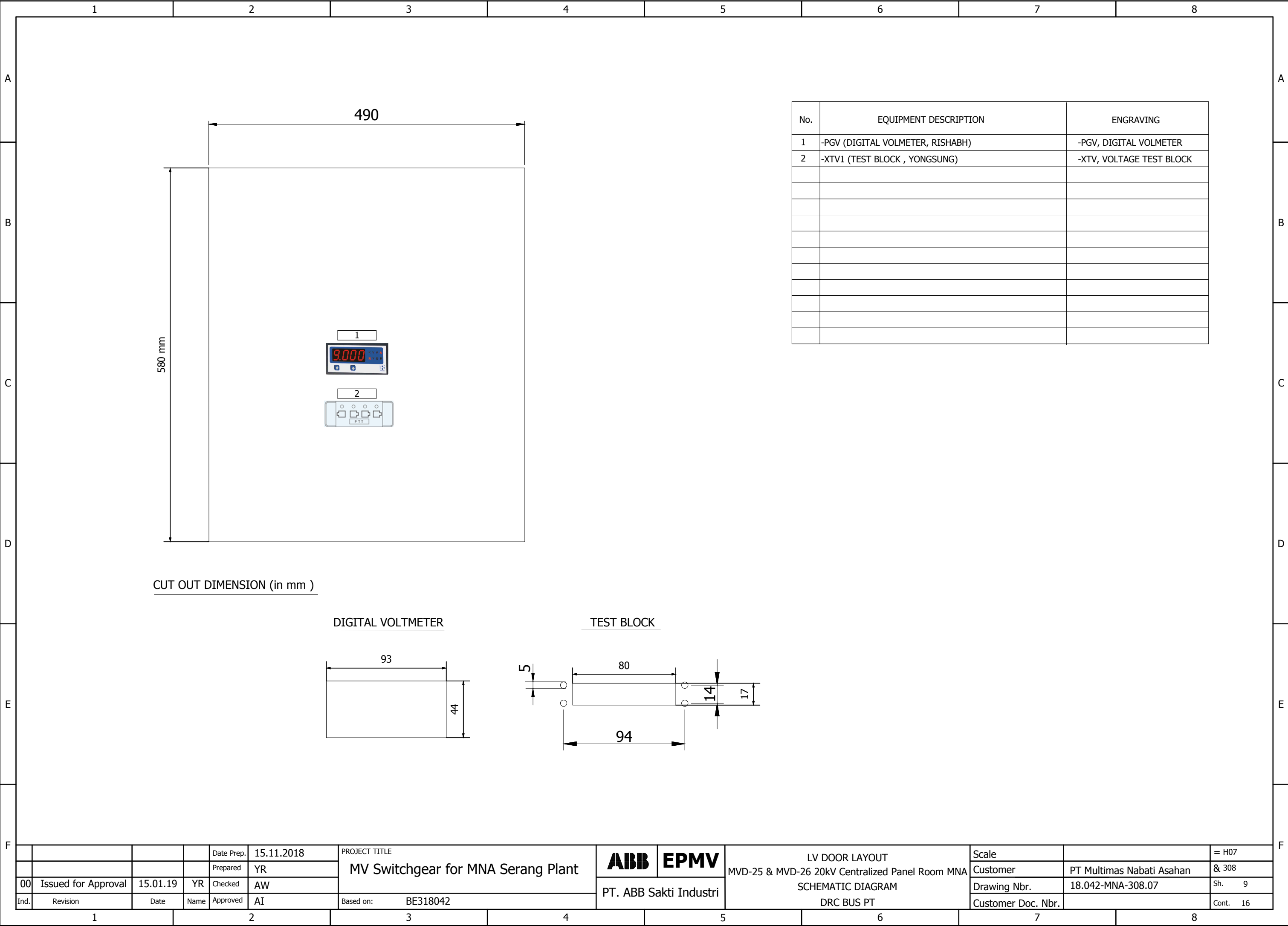


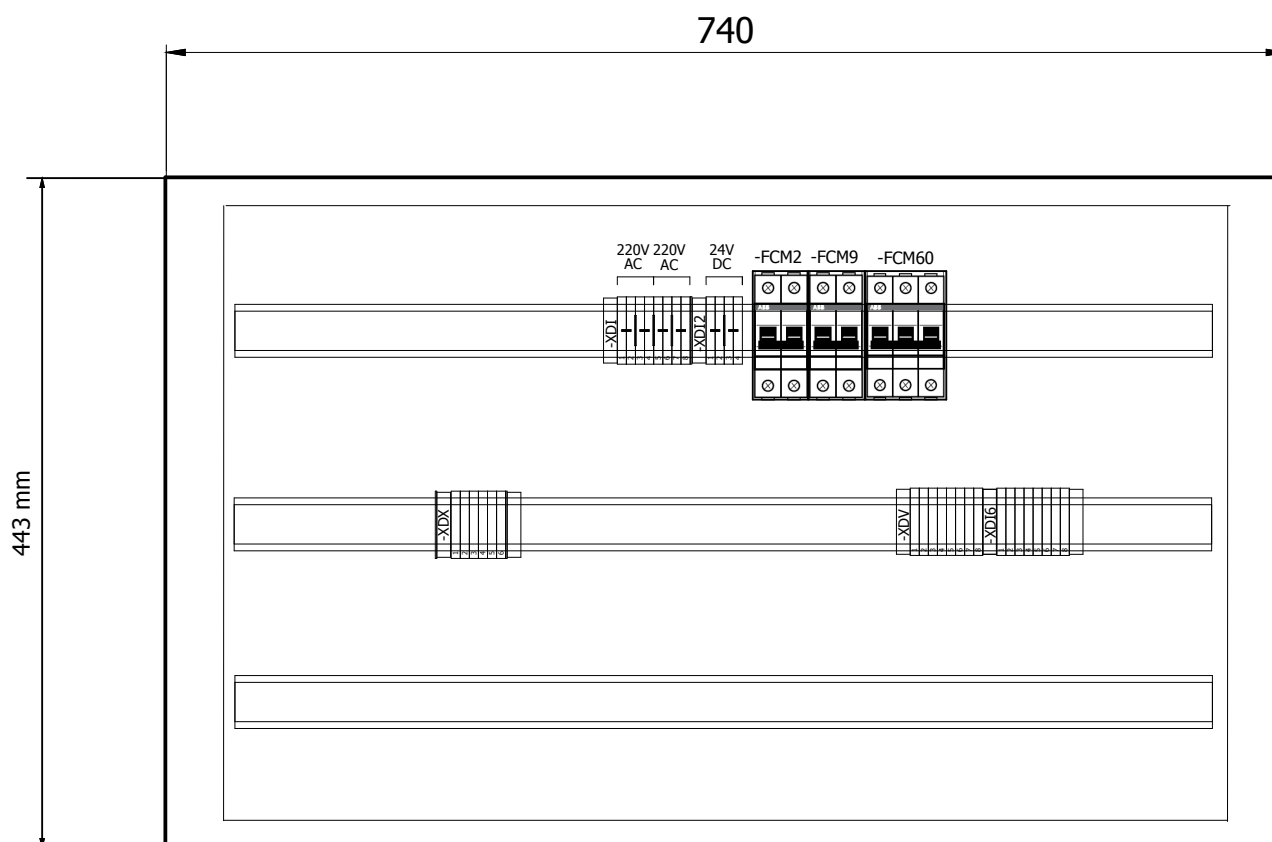
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|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB                | EPMV              | LV BOX LAYOUT<br>MVD-25 & MVD-26 20kV Centralized Panel Room MNA<br>SCHEMATIC DIAGRAM<br>DRC BUS PT | Scale                  |          | = H07                     |
|      |                     |          |      | Prepared   | YR         |                                                         |                    |                   |                                                                                                     | PT. ABB Sakti Industri | Customer | PT Multimas Nabati Asahan |
| 00   | Issued for Approval | 15.01.19 | YR   | Checked    | AW         |                                                         | Drawing Nbr.       | 18.042-MNA-308.07 |                                                                                                     |                        | Sh.      | 10                        |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on: BE318042                                      | Customer Doc. Nbr. |                   |                                                                                                     | Cont.                  | 16       |                           |

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|      |          |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB                    | EPMV               | PART LIST<br><br>MVD-25 & MVD-26 20kV Centralized Panel Room MNA<br><br>SCHEMATIC DIAGRAM<br><br>DRC BUS PT | Scale             |                           | = H07 |
|      |          |          |      | Prepared   | YR         |                                                         |                        |                    |                                                                                                             | Customer          | PT Multimas Nabati Asahan | & 308 |
| 00   |          | 15.01.19 | YR   | Checked    | AW         |                                                         | PT. ABB Sakti Industri | Drawing Nbr.       |                                                                                                             | 18.042-MNA-308.07 | Sh. 11                    |       |
| Ind. | Revision | Date     | Name | Approved   | AI         | Based on: BE318042                                      |                        | Customer Doc. Nbr. |                                                                                                             | Cont. 16          |                           |       |

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